



VCS VERIFICATION REPORT

HEILONGJIANG WANGYUNFENG WIND
POWER PROJECT

(VCS PROJECT ID: 1051)



Document Prepared By

Shenzhen CTI International Certification Co., Ltd

Project Title	Heilongjiang Wangyunfeng Wind Power Project
Version	1.4
Report ID	CTINB-2022-0519-26
Report Title	VCS Verification Report of Heilongjiang Wangyunfeng Wind Power Project
Client	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

Pages	36
Date of Issue	28-Sep-2022
Prepared By	Shenzhen CTI International Certification Co., Ltd
Contact	Address: F8-A CTI Building, No.4 LiuXianSan Road, Xin'an Street, Bao'an District, 518101, Shen Zhen, China Tel: +86 10 65580012 Email: linshunrong@cti-cert.com Website: http://www.cti-cert.org
Approved By	Zhou Lu 
Work Carried Out By	Team Leader: Zhang Lei Team Member: Zhang Wenting Technical Reviewer: Lin Shunrong

Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity "Heilongjiang Wangyunfeng Wind Power Project" (VCS Project ID:1051) for the monitoring period 01/01/2013-31/12/2018 within the 1st VCS crediting period 20/01/2010-19/01/2020, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.2.

The verification was performed on the basis of VCS Standard Version 4.2 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified two CLs, no CAR and no FAR were raised in this monitoring period.

In CTI's opinion, the GHG emission reductions reported for the project in the monitoring report (version 04 dated 28/09/2022) are fairly stated. The GHG emission reductions of the period from 01/01/2013-31/12/2018 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" (version 12.1.0), the monitoring plan contained in the registered VCS PD (i.e. registered CDM PDD version 02 dated 01/08/2011) for the 1st VCS crediting period 20/01/2010-19/01/2020.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and

certified. Request for issuance of VCU's shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the "Heilongjiang Wangyunfeng Wind Power Project" during this monitoring period from 01/01/2013-31/12/2018 amount to 517,149 tCO₂e.

CONTENTS

1 Introduction	6
1.1 Objective	6
1.2 Scope and Criteria	6
1.3 Level of Assurance	7
1.4 Summary Description of the Project	7
2 Verification Process	8
2.1 Method and Criteria	8
2.2 Document Review	8
2.3 Interviews	9
2.4 Site Inspections	10
2.5 Resolution of Findings	10
2.6 Eligibility for Validation Activities	11
3 Validation Findings	11
3.1 Participation under Other GHG Programs	11
3.2 Methodology Deviations	11
3.3 Project Description Deviations	11
3.4 Grouped Project	12
4 Verification Findings	12
4.1 Project Implementation Status	12
4.2 Safeguards	18
4.3 AFOLU-Specific Safeguards	19
4.4 Accuracy of GHG Emission Reduction and Removal Calculations	19
4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals	26
4.6 Non-Permanence Risk Analysis	26
5 Verification conclusion	27
APPENDIX A: Abbreviations	29
APPENDIX B: references	30
APPENDIX C: Competence of team members and technical reviewers	33

APPENDIX D: Corrective Action Requests, clarification requests and Forward Action Requests	36
---	-----------

1 INTRODUCTION

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the “Heilongjiang Wangyunfeng Wind Power Project” (the Project) for the monitoring period 01/01/2013-31/12/2018. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Heilongjiang Wangyunfeng Wind Power Project” for the period 01/01/2013-31/12/2018.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.1)/42/
- VCS Standard (version 4.2)/41/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 12.1.0)/45/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.2, the project size categorization belongs to 'Project' as the expected annual emission reductions is 123,430 tCO₂e lower than 300,000 tCO₂e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.2 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.1)/42/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: wind power project

Project Background

The project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China. The site location's central geographical coordinates are east longitude of 130 ° 39'56'' and north latitude of 47 ° 54'34''. The project activity was registered as a CDM project with Ref.5711 on 31/01/2012 with the crediting period starting from 31/01/2012. The project has been registered as VCS project with Ref. VCS 1051 under VCS Standard Version 3.2 with the start date 20/01/2010 when the commencement of operations activities started as confirmed by checking the operation log/7/, the VCS monitoring report and verification report for the period of 20/01/2010-30/01/2012 /31//32/.

The amount of 195,043 tCO₂e VCU have been issued for the monitoring period 20/01/2010-30/01/2012 within the 1st VCS crediting period. GHG emission reductions have been applied for CDM for CERs within the 1st CDM crediting period, total 72,155 tCO₂e CERs have been issued for the monitoring periods 31/01/2012-31/12/2012.

The installed capacity of the project is 49.3MW equipped with 58 sets of wind generator with unit capacity of 850kW, including 46 sets in type of G52-850 and 12 sets in type of G58-850, manufactured by Gamesa Wind (Tianjin) Co., Ltd.. The electricity generated by the project activity is supplied to Northeast China Power Grid (NECPG), and the project is estimated to deliver 123,430 tCO₂e emission reduction annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (i.e. registered CDM PDD version 02 dated 01/08/2011) and monitoring plan contained in the registered VCS PD, the applied methodology, and the VCS Standard (version 4.2) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader	Zhang Lei	1.1,1.2, 4.1,13.1	☒ DR ☒ SV ☒ RP ☐ TR
Team Member	Zhang Wenting	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered VCS PD /15/, emission reduction calculation spreadsheet/2/, the following documents also were assessed as a part of the verification audit:

- The VCS validation report /31/;
- The previous VER/VCU monitoring report/31/ and verification report/32/, and previous CDM/CER monitoring report /34/and verification report/35/;

- Baseline and monitoring methodology ACM0002 (version 12.1.0) applied by the project/45/;
- Relevant decisions, clarifications and guidance from the Verra/41/-/44/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 20/06/2022, CTI visited the project proponent Hegang Longyuan Wind Power Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /45/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /46/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Wang Xuesong	Station Manager	Hegang Longyuan Wind Power Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement;
Mr. Liu Jiyuan	Operator	Hegang Longyuan Wind Power Co., Ltd.	Calibration of meters and equipment maintenance;
Mr. Fu Hongliang	Operator	Hegang Longyuan Wind Power Co., Ltd.	The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions
Ms. Yu Jinfeng	Local resident	Local Village	The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity;
Mr. Yu Jinhua			
Mr. Wang Wenhua	Staff	Local Environment Protection Bureau	Project's sustainable development contributions.
Mr. Xueqiang	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.	Data collection and ER calculation.

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity sales receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 12.1.0), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR) shall be raised, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 2 CLs and no CAR in this monitoring period. The CLs were satisfactorily addressed by the project proponents in the updated monitoring report version 04 dated 28/09/2022 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/32/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by Bureau Veritas Certification Holding SAS (BVCH) under the CDM rules and registered as CDM project on 31/01/2012, and was registered as VCS project through the GAP validation by DNV Climate Change Services AS under VCS standard version 3.2.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification reports/32/, it is confirmed the verification for this monitoring period 01/01/2013-31/12/2018 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.2.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 31/01/2012 under the UNFCCC with Ref. 5711 with the crediting period starting from 31/01/2012. By checking the UNFCCC website (<https://cdm.unfccc.int/Projects/DB/BVQI1327230197.63/view>), it is confirmed that total 72,155tCO₂e of CERs have been issued for the monitoring periods 31/01/2012-31/12/2012 within the 1st CDM crediting period.

3.2 Methodology Deviations

The validation process /20/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 12.1.0) /45/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the registered VCS PD, the previous VCS monitoring report with the monitoring period 20/01/2010-30/01/2012, the previous CDM verification report with the monitoring period

31/01/2012-31/12/2012, documentation of the project's confirmation of the commissioning and start of operation of the wind turbine, Project construction-start application and permission/8/, Verification report of project construction completion/9/, and through the site visit interview with project owner, it is confirmed that the first turbine started commercial operation on 20/01/2010 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 3.2 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 20/01/2010-30/01/2012 to 20/01/2010-19/01/2020, which lasts for 10 years.

As per VCS requirement, the validation of crediting period renewal was not conducted within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 19/01/2020.

CTI confirms that the project is fully implemented in accordance with the project description, and no any project description deviations. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Heilongjiang Wangyunfeng Wind Power Project” for the monitoring period 01/01/2013-31/12/2018.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China. The central geographical coordinates are verified by GPS system as east longitude of 130°39'56" and north latitude of 47°54'34". The total installed capacity is 49.3MW equipped with 58 sets of wind generator with unit capacity of 850kW, including 46 sets in type of G52-850 and 12 sets in type of G58-850. It was confirmed by checking the operation logs/7/, the validation report, the previous VCS/CDM monitoring reports, verification reports /31-/35/, Project construction-start application and permission/8/ and Verification report of project construction completion/9/, construction start permission of the project was issued on 04/06/2009, the construction contract was signed on 05/06/2009, the first wind turbine started commercial operation on 20/01/2010 and all wind turbines started commercial operation on 12/04/2010. The actual

implementation of the project during this verification period was verified in terms of nameplates of wind turbines /36/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered VCS PD.

The electricity generated by the project activity is supplied to NECPG, which can be confirmed by the Power purchase agreement (PPA) signed between Hegang Longyuan Wind Power Co., Ltd. and Heilongjiang Provincial Power Grid Company /3/. All the monitoring system in operation period is consistent with the description in monitoring plan contained in the registered VCS PD. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the Operation logs/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/46/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the monitoring plan contained in the VCS PD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 12.1.0).

CL 01 was raised for lacking the information of the reference of methodology, which has been successfully closed, Refer to Appendix D for details.

Compliance of monitoring with the monitoring plan

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD/15/, CTI confirms that all parameters stated in the monitoring plan are monitored and reported appropriately. All parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 12.1.0) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex ante” in the MR are checked against the registered VCS PD. The parameter EF_y is the combined margin emissions factor of the grid, which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the VCS PD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and electricity sales receipts;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan, the parameters to be monitored is as below:

$EG_{\text{facility}, y}$	Quantity of net electricity generation supplied by the project to the grid in year y
$EG_{\text{export}, y}$	Electricity exported to the grid by the project
$EG_{\text{import}, y}$	Electricity utilized by the project

Monitored parameters $EG_{\text{facility}, y}$, $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$

By site inspection and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CTI confirmed that the parameter $EG_{\text{facility}, y}$ is the net electricity generation supplied by the project to the grid, which is the difference between the electricity exported to the grid by the project ($EG_{\text{export}, y}$) and the electricity utilized by the project ($EG_{\text{import}, y}$). $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$ are continuously measured by main meter M1 installed at outlet of the main transformer and backup meter M2 installed at the inlet of the main transformer. Both meters are bi-directional and with accuracy of 0.5S. The meter readings of $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$ were continuously measured, daily read and monthly aggregated.

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /7/ and via on-site visit, it is confirmed that both meters were working well during this monitoring period. Via the on-site interview and by checking the operation logs/7/, monthly reading records and electricity sales and purchase receipts confirmed by both the grid company and the project/12/, it is confirmed by the verification team that the cut-off time of electricity transaction is at 24:00 of last day of each month /11/. Month reading records of M1/11/

were used to the report. The electricity sales receipts/12/ issued by the grid company and the project have been used for verifying the records of $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$.

CL 02 was raised for missing two monitored parameters from the monitoring plan, which has been successfully closed, Refer to Appendix D for details.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 12.1.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Meters	Type	SN	Accuracy class	Calibration Date	Validity	Calibration Entity
M1	DTSD719	420052504 0909208867	0.5S	12/05/2012	11/05/2013	Hegang Metrological Verification Institution
				10/05/2013	09/05/2014	
				08/05/2014	07/05/2015	
				06/05/2015	05/05/2016	
				04/05/2016	03/05/2017	
				02/05/2017	01/05/2018	
				30/04/2018	29/04/2019	
				12/05/2012	11/05/2013	
M2	DTSD719	420022204 0909208865	0.5S	10/05/2013	09/05/2014	
				08/05/2014	07/05/2015	
				06/05/2015	05/05/2016	
				04/05/2016	03/05/2017	
				02/05/2017	01/05/2018	
				30/04/2018	29/04/2019	
				12/05/2012	11/05/2013	
				10/05/2013	09/05/2014	

Calibration records and accreditation certificates/16//17/have been verified by the verification team. The actual accuracy of meters M1 and M2 is verified to be 0.5S, which is consistent with the requirement in the monitoring plan (accuracy degree is not less than 0.5%) contained in the registered VCS PD and in compliance with the Technical administrative code of electric energy metering /19/. The calibration frequency of meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan. And the calibrations are verified to be valid for the whole reporting period.

Data management and control

Hegang Longyuan Wind Power Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/46/ and check records /7//11/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the VCS PD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

By checking the UNFCCC website (<https://cdm.unfccc.int/Projects/DB/BVQI1327230197.63/view>), it is confirmed that emission reductions within the monitoring periods 31/01/2012-31/12/2012 have been issued as CERs within the 1st CDM crediting period 31/01/2012-30/01/2019.

Therefore, it is concluded the monitoring periods of the project activity is consecutive and no double counting issuance has occurred for this verified monitoring period 01/01/2013-31/12/2018 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCU's issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction(CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tons of CO₂e/year as per Measures for the Administration of Carbon Emission Trading/23/ including thermal power generation, petrochemical, chemical, building materials,

iron and steel, etc. By checking the business license/18/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/24/.

Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/23/. Also, CCER mechanism has been suspended by Chinese government since 2017/25/, the project activity will not and is not allowed to registered under CCER mechanism. Thus, the assessment team concluded that the project activity not involved on other emissions trading programs and other binding limits.

The statement of no double counting/26/ submitted by PP was checked by the verification team and confirmed to be correct in terms of the regulations and database of CCER and China ETS.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- SDG 7, Indicator 7.2 Renewable energy supplied to the power grid.

The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in NECPG. During this monitoring period, 503,067.772 MWh of net electricity generated by wind energy was delivered to the NECPG. It is verified by checking the daily and monthly reading records/11/ and via on-site inspection. Meanwhile, by checking the approved VCS monitoring report for the first monitoring period/31/, it is confirmed 692,799.292MWh of cumulative renewal electricity verified has been supplied to the NECPG since the project operation until the end of this monitoring period. This contributes to increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030"/39/.

- SDG 8, Indicator 8.5 Numbers of job opportunities provided by implementation of the project.

Through interviews with project proponents, employees and local villagers, verification of employee rosters, per capita wage statements provided by PP, partial labour contracts and statistics published by the national statistics data on disposable income per capita from 2013-2018/27/-/29/, it was confirmed that project activities provided 12 permanent employment opportunities for local residents and provides higher salary from 131,279 RMB/year in 2013 to 200,988 RMB/year in 2018 above the national average line level of 26,467 RMB/year in 2013 and 39,251 RMB/year in 2018. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to one of the China 's actions for promoting sustainable developing, " Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" ;

- SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the daily and monthly reading records/11/, 517,149tCO₂e GHG emission reductions are

generated by the project activity during this monitoring period. And by checking the VCS monitoring report for the first monitoring period/31/, it is confirmed the cumulative emission reductions are 712,192 tCO₂e over the two VCS monitoring periods. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas"/40/.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and wastewater, noise, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the staff roster/27/, the Explanation of the employment offered by the project /28/ and the annual average salary provided by the project/29/, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above urban disposable income per capita. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training/6/ organised by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them. As verified in Section 4.1 above, 503,067.772 MWh of clean energy was supplied by the project activity to local area and 517,149 tCO₂e GHG emission reductions were generated. These contributions achieve the China's stated sustainable development priorities in respects to increasing labor force participation rate, increasing the share of clean energy consumption and strengthening resistance capacity to climate risks/40/. Therefore, the implementation of the project activity is beneficial to local social economic environment.

4.2.2 Local Stakeholder Consultation

The stakeholder symposium was held in Luobei County 06/05/2009. Totally 10 stakeholder representatives participated the symposium, respectively from the Development and Reform Bureau of Luobei County, several other department of local government and some villagers around plant. In order to finish this survey, project owner designed the style of Questionnaire and visited to Damahe Forest Farm, Luobei County to do door to door Interviews before the Symposium. Questionnaires were distributed according to the principle of both representation and randomness in order to reflect the public opinions and comments in a fair and real manner. Totally 40 questionnaires returned out of 40 with 100% response rate.

By checking the the summary of comments/13/ and the validation report for VCS registered PD/31/, it is confirmed that the stakeholder meeting showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, the project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in May 2014, July 2016 and May 2018 during this monitoring period. Via checking the results of the questionnaire/14/, it is confirmed by the verification team that no comments were received. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the grid by the project during year y ($EG_{\text{facility}, y}$) by the validated ex-ante fixed grid emission factor (EF_y).

$$BE_y = EG_{\text{facility}, y} \times EF_y$$

Combined margin CO₂ emission factor for grid connected power generation (EF_y)

EF_y is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 1.0280 tCO₂e/MWh.

The net electricity supplied to the grid by the project ($EG_{\text{facility}, y}$)

As per the Section 4.1 above, $EG_{\text{facility}, y}$ is determined as the differences between electricity exported to and imported from the grid.

The verification team has checked the monthly reading records, electricity receipts, and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and $EG_{\text{facility}, y}$ calculation are summarized in Table 1 as follow:

Table 1 The net electricity supplied to the grid by the Project (EG_{facility, y})

Period	EG _{export,y}			EG _{import,y}			EG _{facility,y} (MWh)
	Monthly Recording from main meter (M1) (MWh)	Sales Receipt (MWh)	Conservative Quantity of the electricity exported to the grid (MWh)	Monthly Recording from main meter (M1) (MWh)	Purchase Receipt (MWh)	Conservative Quantity of the electricity imported from the grid (MWh)	
	A	B	C=min(A,B)	D	E	F=max(D,E)	
01/01/2013-31/01/2013	11,354.640	11,354.640	11,354.640	17.688	17.688	17.688	11,336.952
01/02/2013-28/02/2013	9,191.160	9,191.160	9,191.160	66.000	66.000	66.000	9,125.160
01/03/2013-31/03/2013	10,307.880	10,307.880	10,307.880	34.452	34.452	34.452	10,273.428
01/04/2013-30/04/2013	7,316.760	7,316.760	7,316.760	14.652	14.652	14.652	7,302.108
01/05/2013-31/05/2013	5,825.160	5,825.160	5,825.160	18.084	18.084	18.084	5,807.076
01/06/2013-30/06/2013	4,877.400	4,877.400	4,877.400	31.020	31.020	31.020	4,846.380
01/07/2013-31/07/2013	6,316.200	6,316.200	6,316.200	28.116	28.116	28.116	6,288.084
01/08/2013-31/08/2013	3,033.360	3,033.360	3,033.360	26.004	26.004	26.004	3,007.356
01/09/2013-30/09/2013	7,710.120	7,710.120	7,710.120	40.788	40.788	40.788	7,669.332
01/10/2013-31/10/2013	10,178.520	10,178.520	10,178.520	11.220	11.220	11.220	10,167.300
01/11/2013-30/11/2013	9,617.520	9,617.520	9,617.520	31.020	31.020	31.020	9,586.500
01/12/2013-31/12/2013	9,514.560	9,514.560	9,514.560	39.996	39.996	39.996	9,474.564
Subtotal	95,243.280	95,243.280	95,243.280	359.040	359.040	359.040	94,884.240
01/01/2014-31/01/2014	9,554.160	9,554.160	9,554.160	26.928	26.928	26.928	9,527.232
01/02/2014-28/02/2014	9,410.280	9,410.280	9,410.280	26.400	26.400	26.400	9,383.880
01/03/2014-31/03/2014	7,227.000	7,227.000	7,227.000	38.808	38.808	38.808	7,188.192
01/04/2014-30/04/2014	6,974.880	6,974.880	6,974.880	18.216	18.216	18.216	6,956.664

01/05/2014-31/05/2014	8,364.840	8,364.840	8,364.840	22.176	22.176	22.176	8,342.664
01/06/2014-30/06/2014	3,692.040	3,692.040	3,692.040	18.480	18.480	18.480	3,673.560
01/07/2014-31/07/2014	5,095.200	5,095.200	5,095.200	46.728	46.728	46.728	5,048.472
01/08/2014-31/08/2014	3,923.040	3,923.040	3,923.040	22.176	22.176	22.176	3,900.864
01/09/2014-30/09/2014	6,085.200	6,085.200	6,085.200	25.344	25.344	25.344	6,059.856
01/10/2014-31/10/2014	9,768.000	9,768.000	9,768.000	14.520	14.520	14.520	9,753.480
01/11/2014-30/11/2014	9,580.560	9,580.560	9,580.560	10.560	10.560	10.560	9,570.000
01/12/2014-31/12/2014	9,005.040	9,005.040	9,005.040	36.960	36.960	36.960	8,968.080
Subtotal	88,680.240	88,680.240	88,680.240	307.296	307.296	307.296	88,372.944
01/01/2015-31/01/2015	11,201.520	11,201.520	11,201.520	10.560	10.560	10.560	11,190.960
01/02/2015-28/02/2015	4,052.400	4,052.400	4,052.400	88.440	88.440	88.440	3,963.960
01/03/2015-31/03/2015	3,711.810	3,711.810	3,711.810	80.520	80.520	80.520	3,631.290
01/04/2015-30/04/2015	3,247.200	3,247.200	3,247.200	26.400	26.400	26.400	3,220.800
01/05/2015-31/05/2015	7,459.320	7,459.320	7,459.320	10.560	10.560	10.560	7,448.760
01/06/2015-30/06/2015	4,585.680	4,585.680	4,585.680	26.400	26.400	26.400	4,559.280
01/07/2015-31/07/2015	2,873.610	2,873.610	2,873.610	34.320	34.320	34.320	2,839.290
01/08/2015-31/08/2015	3,574.560	3,574.560	3,574.560	44.880	44.880	44.880	3,529.680
01/09/2015-30/09/2015	5,062.200	5,062.200	5,062.200	30.360	30.360	30.360	5,031.840
01/10/2015-31/10/2015	4,477.440	4,477.440	4,477.440	14.520	14.520	14.520	4,462.920
01/11/2015-30/11/2015	5,446.680	5,446.680	5,446.680	0.000	0.000	0.000	5,446.680
01/12/2015-31/12/2015	7,171.560	7,171.560	7,171.560	59.400	59.400	59.400	7,112.160
Subtotal	62,863.980	62,863.980	62,863.980	426.360	426.360	426.360	62,437.620
01/01/2016-31/01/2016	7,962.240	7,962.240	7,962.240	42.240	42.240	42.240	7,920.000
01/02/2016-29/02/2016	3,157.440	3,157.440	3,157.440	19.800	19.800	19.800	3,137.640

01/03/2016-31/03/2016	5,260.200	5,260.200	5,260.200	46.200	46.200	46.200	5,214.000
01/04/2016-30/04/2016	6,214.560	6,214.560	6,214.560	19.800	19.800	19.800	6,194.760
01/05/2016-31/05/2016	10,750.080	10,750.080	10,750.080	10.560	10.560	10.560	10,739.520
01/06/2016-30/06/2016	4,366.560	4,366.560	4,366.560	25.080	25.080	25.080	4,341.480
01/07/2016-31/07/2016	4,345.440	4,345.440	4,345.440	34.320	34.320	34.320	4,311.120
01/08/2016-31/08/2016	5,530.800	5,530.800	5,530.800	31.680	31.680	31.680	5,499.120
01/09/2016-30/09/2016	6,733.320	6,733.320	6,733.320	29.040	29.040	29.040	6,704.280
01/10/2016-31/10/2016	12,475.320	12,475.320	12,475.320	5.676	5.676	5.676	12,469.644
01/11/2016-30/11/2016	8,271.120	8,271.120	8,271.120	21.384	21.384	21.384	8,249.736
01/12/2016-31/12/2016	7,906.800	7,906.800	7,906.800	11.220	11.220	11.220	7,895.580
Subtotal	82,973.880	82,973.880	82,973.880	297.000	297.000	297.000	82,676.880
01/01/2017-31/01/2017	7,398.600	7,398.600	7,398.600	25.344	25.344	25.344	7,373.256
01/02/2017-28/02/2017	4,608.120	4,608.120	4,608.120	14.388	14.388	14.388	4,593.732
01/03/2017-31/03/2017	4,264.920	4,264.920	4,264.920	20.856	20.856	20.856	4,244.064
01/04/2017-30/04/2017	9,212.280	9,212.280	9,212.280	15.708	15.708	15.708	9,196.572
01/05/2017-31/05/2017	8,466.480	8,466.480	8,466.480	7.496	7.496	7.496	8,458.984
01/06/2017-30/06/2017	4,509.120	4,509.120	4,509.120	29.568	29.568	29.568	4,479.552
01/07/2017-31/07/2017	5,058.240	5,058.240	5,058.240	27.456	27.456	27.456	5,030.784
01/08/2017-31/08/2017	5,732.760	5,732.760	5,732.760	40.788	40.788	40.788	5,691.972
01/09/2017-30/09/2017	6,623.760	6,623.760	6,623.760	26.004	26.004	26.004	6,597.756
01/10/2017-31/10/2017	11,201.520	11,201.520	11,201.520	16.632	16.632	16.632	11,184.888
01/11/2017-30/11/2017	12,112.320	12,112.320	12,112.320	13.992	13.992	13.992	12,098.328
01/12/2017-31/12/2017	6,358.440	6,358.440	6,358.440	18.480	18.480	18.480	6,339.960
Subtotal	85,546.560	85,546.560	85,546.560	256.712	256.712	256.712	85,289.848

01/01/2018-31/01/2018	11,052.360	11,052.360	11,052.360	48.840	48.840	48.840	11,003.520
01/02/2018-28/02/2018	6,864.000	6,864.000	6,864.000	40.920	40.920	40.920	6,823.080
01/03/2018-31/03/2018	10,858.320	10,858.320	10,858.320	23.760	23.760	23.760	10,834.560
01/04/2018-30/04/2018	7,357.680	7,357.680	7,357.680	6.600	6.600	6.600	7,351.080
01/05/2018-31/05/2018	5,030.520	5,030.520	5,030.520	21.120	21.120	21.120	5,009.400
01/06/2018-30/06/2018	5,433.120	5,433.120	5,433.120	19.800	19.800	19.800	5,413.320
01/07/2018-31/07/2018	4,534.200	4,534.200	4,534.200	19.800	19.800	19.800	4,514.400
01/08/2018-31/08/2018	3,288.120	3,288.120	3,288.120	38.280	38.280	38.280	3,249.840
01/09/2018-30/09/2018	7,117.440	7,117.440	7,117.440	22.440	22.440	22.440	7,095.000
01/10/2018-31/10/2018	7,897.560	7,897.560	7,897.560	19.800	19.800	19.800	7,877.760
01/11/2018-30/11/2018	8,235.480	8,235.480	8,235.480	11.880	11.880	11.880	8,223.600
01/12/2018-31/12/2018	12,029.160	12,029.160	12,029.160	18.480	18.480	18.480	12,010.680
Subtotal	89,697.960	89,697.960	89,697.960	291.720	291.720	291.720	89,406.240
Total	505,005.900	505,005.900	505,005.900	1,938.128	1,938.128	1,938.128	503,067.772

The corresponding baseline emission reductions during this monitoring period is calculated as:

Year	EG _{facility, y} (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
01/01/2013-31/12/2013	94,884.240	1.0280	97,540
01/01/2014-31/12/2014	88,372.944	1.0280	90,847
01/01/2015-31/12/2015	62,437.620	1.0280	64,185
01/01/2016-31/12/2016	82,676.880	1.0280	84,991
01/01/2017-31/12/2017	85,289.848	1.0280	87,677
01/01/2018-31/12/2018	89,406.240	1.0280	91,909
Total	503,067.772	-	517,149

Project emissions

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from Heilongjiang Wangyunfeng Wind Power Project is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 12.1.0).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO ₂ e)
01/01/2013-31/12/2013	97,540
01/01/2014-31/12/2014	90,847
01/01/2015-31/12/2015	64,185
01/01/2016-31/12/2016	84,991
01/01/2017-31/12/2017	87,677
01/01/2018-31/12/2018	91,909
Total ERs claimed (in 2,191 days)	517,149

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered CDM-PDD

The emission reductions claimed are 517,149 tCO₂e in this monitoring period (i.e. 2,191 days). Compared with expected emission reductions 740,918 tCO₂e (calculated as 123,430tCO₂/365d*2,191d), the reported emission reductions in this monitoring period are 30.20% less than the expected in the registered VCS PD. Considering the natural fluctuate of the wind resource, actual generator operation conditions and limitation of the local grid absorption capacity, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

Therefore, CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, sales receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CTI concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Heilongjiang Wangyunfeng Wind Power Project” in China (VCS Project ID: 1051) for the period 01/01/2013-31/12/2018.

The verification is based on the baseline and monitoring methodology ACM0002 (version 12.1.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard Version 4.2 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.2 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Heilongjiang Wangyunfeng Wind Power Project" for the period 01/01/2013-31/12/2018 are fairly stated in the monitoring report (version 04 dated 28/09/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 12.1.0) and the monitoring plan contained in the registered VCS PD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 01/01/2013-31/12/2018

Verified GHG emission reductions and removals in the above verification period:

<i>Period</i>	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01/01/2013-31/12/2013	97,540	0	0	97,540
01/01/2014-31/12/2014	90,847	0	0	90,847
01/01/2015-31/12/2015	64,185	0	0	64,185
01/01/2016-31/12/2016	84,991	0	0	84,991
01/01/2017-31/12/2017	87,677	0	0	87,677
01/01/2018-31/12/2018	91,909	0	0	91,909
Total	517,149	0	0	517,149

Zhang Lei

Mr.Zhang Lei

Technical Leader

28-Sep-2022

Shunrong Lin

Ms. Lin Shunrong

Team Reviewer

28-Sep-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
ER	Emission Reduction
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MP	MONITORING REPORT
MR	Monitoring Report
NECPG	Northeast China Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: VER/VCU Monitoring Report for Heilongjiang Wangyunfeng Wind Power Project, V01 dated 18/06/2022, version 02 dated 22/06/2022, version 03 dated 19/08/2022, version 04 dated 28/09/2022
/2/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Emission reduction calculation spreadsheet for Heilongjiang Wangyunfeng Wind Power Project, V01 dated 18/06/2022
/3/	Hegang Longyuan Wind Power Co., Ltd. and Heilongjiang Provincial Power Grid Company: Power Purchase Agreement for Heilongjiang Provincial Power Grid Company
/4/	Hegang Longyuan Wind Power Co., Ltd.: Diagram of power connection system
/5/	Hegang Longyuan Wind Power Co., Ltd.: VER monitoring manual and management procedure.
/6/	Hegang Longyuan Wind Power Co., Ltd.: Records of training for on-site staff.
/7/	Hegang Longyuan Wind Power Co., Ltd.: Operation logs for project activity, during the period of 20/01/2010-19/01/2020.
/8/	Hegang Longyuan Wind Power Co., Ltd. and Helongjiang Runhua Electric Power Engineering Project Supervision Co., Ltd: Project construction-start application and permission, dated 04/06/2009.
/9/	Zhongrui Yuehua Certified Public Accountants: Verification report of project construction completion of Heilongjiang Wangyunfeng Wind Power Project, dated 31/10/2010.
/10/	Project construction-start application and permission, dated 4 June 2009.
/11/	Hegang Longyuan Wind Power Co., Ltd.: Monthly reading records of Heilongjiang Wangyunfeng Wind Power Project from 01/01/2013-31/12/2018
/12/	Heilongjiang Provincial Power Grid Company.: Electricity sales receipts for electricity exported to and imported from the grid by the project activity for 01/01/2013-31/12/2018
/13/	Meeting memo of the stakeholder consultation and the participants' signature form, 26/07/2006
/14/	Questionnaires for local stakeholders

/15/	Hegang Longyuan Wind Power Co., Ltd.: Registered CDM PDD (version 02 dated 01/08/2011)
/16/	Heilongjiang Provincial Market Supervision Administration: Accreditation certificate of Hegang Metrological Verification Institution, issued on 22/06/2012, valid up to 21/06/2016; issued on 22/06/2016, valid up to 21/06/2020; issued on 22/06/2016, valid up to 21/06/2024
/17/	Hegang Metrological Verification Institution: Calibration certificates for meters M1 and M2 covering this monitoring period
/18/	Business license of Hegang Longyuan Wind Power Co., Ltd.
/19/	State Economic and Trade Commission: Technical administrative code of electric energy metering (JJG596-2012)
/20/	UNFCCC, http://cdm.unfccc.int
/21/	Gold Standard, http://www.goldstandard.org/
/22/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/23/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/24/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/25/	Announcement of the National Development and Reform Commission, Code 2 in 2017, issued by National Development and Reform Commission, dated 03/14/2017
/26/	Hegang Longyuan Wind Power Co., Ltd.: Statement of no double counting
/27/	Hegang Longyuan Wind Power Co., Ltd.: Staff roster
/28/	Hegang Longyuan Wind Power Co., Ltd.: Explanatory material on the number of people employed and the average salary provided by the project.
/29/	Statistical Bulletin on National Economic and Social Development by Year: http://www.stats.gov.cn/search/
/30/	DNV Climate Change Services AS: VCS Validation Report (included in verification report no. 2013-9027, dated 23/04/2013)
/31/	VER/VCU Monitoring Report for Heilongjiang Wangyunfeng Wind Power Project of previous monitoring period 20/01/2010-30/01/2012
/32/	DNV Climate Change Services AS: VCS Verification Report for Heilongjiang Wangyunfeng Wind Power Project of previous monitoring period 20/01/2010-30/01/2012

/33/	Bureau Veritas Certification Holding SAS (BVCH): CDM Validation Report, CHINAVAl/0231/2009, revision 01 dated 04/01/2012
/34/	CDM/CER Monitoring Reports for Heilongjiang Wangyunfeng Wind Power Project of previous CDM monitoring periods 31/01/2012-31/12/2012
/35/	CDM Verification Reports for Heilongjiang Wangyunfeng Wind Power Project of previous monitoring period 31/01/2012-31/12/2012
/36/	Gamesa Wind (Tianjin) Co., Ltd.: Nameplates of the equipment
/37/	Feasibility Study Report of Heilongjiang Wangyunfeng Wind Power Project
/38/	Co-signed by the Gamesa Wind (Tianjin) Co., Ltd. and Hegang Longyuan Wind Power Co., Ltd.: Technical agreement of wind turbines purchase contract for the project activity Heilongjiang Wangyunfeng Wind Power Project, dated 12/03/2009
/39/	Action Plan for Carbon Dioxide Peaking Before 2030 published by the State Council on 24/10/2021
/40/	China's National Programme to Address Climate Change published by the State Council on 28/03/2008

Methodologies, tools and other guidance

/41/	Verified Carbon Standard: VCS Standard, version 4.2
/42/	Verified Carbon Standard: VCS Program Guide, version 4.1
/43/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/44/	Verified Carbon Standard: Registration and Issuance Process, version 4.1
/45/	UNFCCC EB: Approved methodology, ACM0002 , version 12.1.0

Persons interviewed

/46/	Mr. Wang Xuesong, Station Manager, Hegang Longyuan Wind Power Co., Ltd. Mr. Liu Jiyuan, Operator, Hegang Longyuan Wind Power Co., Ltd. Mr. Fu Hongliang, Operator, Hegang Longyuan Wind Power Co., Ltd. Ms. Yu Jinfeng, Local resident, Local Village Mr. Yu Jinhua, Local resident, Local Village Mr. Wang Wenhua, Staff, Local Environment Protection Bureau Ms. Xueqiang, Project Manager, Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
------	--

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Lei ZHANG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation
	TA 1.2: Energy generation from renewable energy sources
SS 4: Manufacturing industries	TA 4.1: Cement and lime production
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

Ms. Wenting ZHANG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	-	-	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN

Wu Lin

Technical Competent Manager

Shenzhen, 01/03/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	It is requested to clarify the reference of the methodology.	The reference of the methodology. had been updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 01 is closed.
CL 02	Section 4.2 of MR (V01 dated 18/06/2022) missing two monitored parameters from the monitoring plan. It is requested to clarify the monitored parameters $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$.	The monitored parameters $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$ had been updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 02 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA